

Project: L'ORÉAL

Feature: Free Location Search Rule Based on Article Classification and Dynamic Picking Threshold

Version: 1.1

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◆ **DOCUMENT METADATA (for Vector DB)**

```
{  
  "document_type": "Technical Specification",  
  "project": "L'OREAL",  
  "system": "Speed WMS",  
  "module": "Location Search",  
  "feature": "Free Location Search Rule",  
  "search_rule_code": "0003",  
  "related_rules": "RG0002 (ZPA search)",  
  "activity_parameters": ["ART_ALPHA3", "ART_ALPHA10", "ART_ALPHA11",  
    "ART_ALPHA8", "ART_ALPHA9"],  
  "version": "1.1"  
}
```

◆ **SECTION 1 — Business Context**

1.1 Functional Context

Within the L'Oréal project, several location search rules are defined to meet storage constraints.

These rules are based on article classification criteria.

The client now wants to configure a Free Location Search rule that:

- Counts the number of locations containing a specific article within a defined zone.

- Applies decision logic based on:
 - Article Division (ART_ALPHA11 – Level 2 classification)
 - Aerosol level (ART_ALPHA3)
 - Dynamic picking threshold (ART_ALPHA10 – ART_NUM2)
-

1.2 Objective of the Rule

The rule must:

1. Count the number of locations containing the same article within a given zone.
2. Compare that count to the article's dynamic picking threshold (ART_ALPHA10).
3. Apply the following logic:

Case 1 — Location Count < Threshold

→ Assign a new free location in the defined zone.

Case 2 — Location Count ≥ Threshold

→ Do not propose a new location.

→ Move to the next storage sequence (e.g., reserve storage).

◆ SECTION 2 — Example Scenarios

Example 1 — Aerosol Picking Zone

Conditions:

- Article classification level 2 (ART_ALPHA11) = 20
- ART_ALPHA3 (Aerosol Level) IN (1, 2, 3)
- Support quality code (QUA_CODE) = 'RC01'
- Number of locations containing the article in zone LO_P_CPD_AER < ART_ALPHA10 (article threshold)

Then:

- Application criteria are validated
 - Assign a new free location in zone LO_P_CPD_AER
-

Example 2 — PLV Zone

Conditions:

- QUA_CODE IN ('RC01', 'RL01', 'RA01', 'RP01')
- ART_ALPHA9 IN ('101','102')
- FAR_CODE IN (20, 30)
- ART_ALPHA11 IN (20, 30)
- Number of locations containing the article in zone LO_P_ALL_PLV < ART_ALPHA10 (article threshold)

Then:

- Application criteria are validated
 - Assign a new free location in zone LO_P_ALL_PLV
-

◆ SECTION 3 — Article Parameter Configuration

Before implementing the rule, article IC parameters must be configured:

Parameter	Purpose
ART_ALPHA3	Aerosol Level
ART_ALPHA10 (ART_NUM2)	Dynamic Picking Threshold
ART_ALPHA11	Article Family / Division

These parameters drive the rule logic.

◆ SECTION 4 — Technical Design

4.1 Search Rule Creation

Create a new free search rule:

Search Rule Code: 0003

This rule must be based on:

- The 12 existing rules in RG0002 (ZPA search).
-

4.2 Required Parameters in the Free Search Rule

The rule must support the following filters:

1 Quality Code (QUA_CODE)

- IN list parameter
 - NOT IN list parameter
-

2 ART_ALPHA3 (Aerosol Level)

- IN list parameter
 - NOT IN list parameter
-

3 Threshold (ART_NUM2 from ART_ALPHA10)

- Retrieved from article IC
 - Used to compare with number of locations
-

4 Article Family (ART_ALPHA11)

- IN list parameter
 - NOT IN list parameter
-

5 ART_ALPHA8

- IN list parameter
 - NOT IN list parameter
-

6 ART_ALPHA9

- IN list parameter
 - NOT IN list parameter
-

7 Zone Name

- Hardcoded (one single zone per rule)
-

8 Support Template (Gabarit Support)

- Hardcoded
 - Based on ZPA search rules logic
-

◆ SECTION 5 — Core Logic

Threshold Control

location_count = COUNT_LOCATIONS(article, zone)

threshold = ARTICLE.ART_NUM2

IF location_count >= threshold THEN

RETURN NO_LOCATION

ELSE

RETURN FREE_LOCATION_IN_ZONE

ENDIF

Behavior

- If threshold reached → do not propose location
 - If threshold not reached → propose free location in defined zone
-

◆ SECTION 6 — Functional Summary (RAG-Optimized)

What is the purpose?

Limit number of picking locations per article in a zone.

What controls the maximum number?

ART_ALPHA10 (ART_NUM2 – Dynamic Picking Threshold)

What classification fields are used?

- **ART_ALPHA3 (Aerosol Level)**
- **ART_ALPHA11 (Division)**
- **ART_ALPHA8**
- **ART_ALPHA9**

What happens if threshold is reached?

No new location is proposed.

What happens if threshold is not reached?

A free location in the zone is assigned.

Which rule is created?

Search Rule 0003

What existing logic is reused?

RG0002 (ZPA search)

◆ **SECTION 7 — Client Validation**

AGL acknowledges receipt and validates all elements included in this document.

BK Systèmes commits to delivering a solution that meets the validated requirements within the scope of the approved quotation.

Any additional request or evolution beyond this document may require additional quotation.

Any provision not explicitly stated in this document is not covered and remains at the discretion of the concerned parties.

Signed at Vaulx-Milieu, on 25/03/2025.

Company stamp and signature: